

Instructor Key: Day 7 Mastery Quiz: Ordered Validation

if/elif/else order to first-true route evidence

Instructor key. Same layout as student quiz; solutions filled in response boxes. Print duplex on one sheet.

Name		Section		Date	
Score	_____/10		Mastery check		secure / developing / needs support

Engineering situation and quiz rule

The checkout route now has several possible holds. Python checks the rules in order and stops at the first true route. Trace the order before choosing the final route.

1 Reference code for Items 1–10

[trace](#) [explain](#)

```
1 damage_note = ""
2 charge_percent = 83
3 calibration_label = "expired"
4 borrower_initials = "MK"
5
6 if damage_note != "":
7     route = "repair bench"
8 elif charge_percent < 80:
9     route = "charge station"
10 elif calibration_label != "current":
11     route = "calibration bench"
12 elif borrower_initials == "":
13     route = "borrower log needed"
14 else:
15     route = "release kit"
```

2 Items 1–5: trace the route evidence

[predict](#) [route](#)

Pt.	Prompt	My response
1	What is the first condition Python checks? Is it true or false here?	Key: <code>damage_note != ""</code> is checked first; it is false because <code>damage_note</code> is blank.
1	What is the second condition Python checks? Is it true or false here?	Key: <code>charge_percent < 80</code> is checked second; <code>83 < 80</code> is false.
1	What is the third condition Python checks? Is it true or false here?	Key: <code>calibration_label != "current"</code> is true because the label is expired.
1	What final route is assigned?	Key: <code>"calibration bench"</code> .
1	Which later condition is skipped after the first true condition is found?	Key: <code>borrower_initials == ""</code> is skipped after the calibration rule is true.

Pt.	Prompt	My response
1	If damage_note were "cracked case", what route would be assigned even though calibration is expired?	Key: "repair bench" because damage is the first true route and stops the chain.
1	If charge_percent were 75 and calibration_label were expired, what route would be assigned?	Key: "charge station" because low charge appears before calibration.
1	If all hold conditions were false, what route would be assigned?	Key: "release kit".
1	Explain why ordered validation is different from checking every rule independently.	Key: In if/elif/else, the chain stops at first true; independent checks could record multiple issues.
1	Write one first-true evidence sentence for the original case.	Key: Secure answer names false damage, false charge, true expired calibration, route calibration bench, and skipped later checks.

Support route, not scored

Do not jump to the most interesting problem. Trace the conditions in written order and stop at the first true rule.